

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY,
1st Semester of B. Pharm. Examination
University Theory Examination April-2016

MA 131: Basic Mathematics

Date: 20/04/2016, Wednesday

Time: 10:00a.m to 01:00p.m

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION-I

Q-1. Attempt any TWO: [20]

(1)(a) Evaluate : (i) $\lim_{x \rightarrow 3} \frac{2x^2 + 3x - 1}{x^3 - 7}$ (ii) $\lim_{x \rightarrow -1} x^2 + 3x + 2$ [05]

(b) Evaluate: $\int (4 \sec x + 3x^2 + \cos^2 x - 9 \sin x - e^{2x}) dx$. [05]

(2)(a) Find derivative of $f(x) = 3 \cos x \log x + x^2 \sin x$ with respect to x . [05]

(b) Let $U = \{1, 2, 3, 4, 5, 6, 8, 12, 17, 18\}$, $A = \{1, 2, 3, 4, 5, 6\}$, $B = \{3, 6, 8, 12, 17, 18\}$ and $C = \{2, 3, 5, 6, 12, 17\}$. Then find (i) $A \cup B$ (ii) $A \cap C$ (iii) $C - B$ [05]

(iv) $(A \cup B) \cup C$ (v) $(A \cap B)' = A' \cup B'$.

(3)(a) Define One-to-one and Onto function with suitable example. [05]

(b) Find derivative of $f(x) = x^{12} + \cos 6x + e^{5x} + \log 7x + 200 \cos x$ with respect to x . [05]

Q-2 Attempt any TWO: [20]

(1)(a) Evaluate : $\int (x \sin x + x^2 e^x) dx$. [05]

(b) If $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$, $C = \{1, 3, 4\}$ and $D = \{2, 4, 5\}$, verify that [05]
 $(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$.

(2)(a) Write the following sets in property form: [05]

(i) $A = \{1, 3, 5, 7, 9, 11, \dots\}$ (ii) $B = \{1, 4, 9, 16, 25, 36, \dots\}$

(iii) $C = \{a, e, i, o, u\}$ (iv) $D = \{5, 6, 7, 8, 9, 10, 11, 12, 13, 14\}$

(v) $E = \{8, 27, 64, 125, 216\}$

(b) Evaluate: $\lim_{x \rightarrow 4} \frac{x^2 - 5x + 4}{x - 4}$ [05]

(3)(a) Find the range of the function $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = x$ and also draw the graph of the function. [05]

(b) Find derivative of $f(x) = \sin(\log x) + \cot(x^3) + e^{x^2}$. [05]

SECTION-II

Q-3 Attempt any TWO:

[20]

- (1)(a) Find $A+B$ and $A-B$, where $A = \begin{bmatrix} 2 & 1 \\ -7 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} -3 & 0 \\ 4 & 2 \end{bmatrix}$ [05]

- (b) Find the root of the equation (i) $x^2 - 5x + 6 = 0$ (ii) $(x-2)^3 = 0$ [05]

- (2)(a) Find the value of following determinants: [05]

(i) $\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 0 & 0 & 7 \end{vmatrix}$ (ii) $\begin{vmatrix} \sin \theta & \cos \theta \\ -\cos \theta & \sin \theta \end{vmatrix}$

- (b) Tuberculin test reaction of 10 patients is arranged in ascending order after being measured in mm. Find the mean and median size of reaction. 3, 5, 7, 7, 8, 8, 9, 10, 11, 12 [05]

- (3)(a) Solve the following simultaneous equation using Cramer's rule: [05]
 $x + 2y = 9$, $2x - 3y = 4$

- (b) Define median and find the median of following data [05]

Class limits	0-30	30-60	60-90	90-120	120-150	150-180
Frequency	8	13	22	27	18	7

Q-4 Attempt any TWO:

[20]

- (1)(a) Find all minors of below matrix: [05]

$$\begin{bmatrix} 0 & 1 & 3 \\ 2 & 6 & 9 \\ 5 & 3 & 4 \end{bmatrix}$$

- (b) Find the root of the equation (i) $5x^2 - 2x - 4 = 0$ (ii) $x^2 + 1 = 0$. [05]

- (2)(a) Define Diagonal Matrix, Symmetric Matrix, Upper Triangular Matrix with suitable example. [05]

- (b) Calculate the mean deviation about the median for following distribution: [05]

Class limits	3-5	5-7	7-9	9-11	11-13	13-15	15-17
Frequency	5	8	30	82	45	24	6

- (3)(a) Solve the following simultaneous equations using Cramer's rule [05]
 $x + 2y + 3z = 5$, $2x - 3y - z = 3$, $-3x + 4y + 5z = 3$

- (b) Find the mean and median of 10, 23, 18, 38, 65, 92, 40, 58. [05]